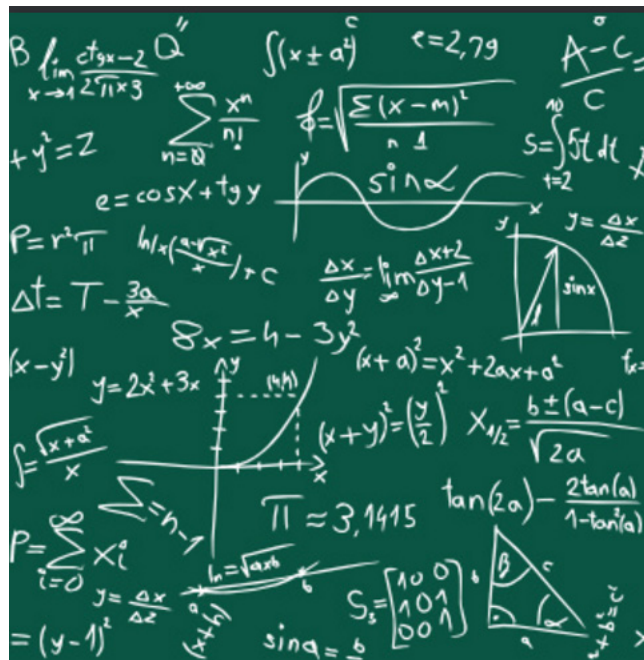


Sequences and limits of sequences



- **Author:** Fabio Furini.
- **Book:** “Analisi matematica 1”, Marco Bramanti, Carlo D. Pagni and Sandro Salsa, Zanichelli.

Contents

1	Definition of sequence and properties	2
1.1	Convergent sequences and definition of the limit of a sequence	6
1.2	Divergent sequences and irregular sequences	14
2	Unbounded sets and their suprema/infima	17
3	Infinitesimal and infinite sequences	18
4	Monotone sequences	21
5	Limits of geometric progressions	24

1 Definition of sequence and properties

- Consider the set $\mathbb{N}$ of non-negative integers, ordered according to the natural order

$$\mathbb{N}: 0, 1, 2, 3, \dots, n, \dots$$

Definition 1: Sequence

A **sequence** is a relation that associates with each natural number $n \in \mathbb{N}$ (or from a certain natural number n_0 onward) a real number $a_n \in \mathbb{R}$.

- A sequence is therefore a **function**:

$$f: \mathbb{N} \rightarrow \mathbb{R}$$

$$f: n \mapsto a_n$$

or possibly

$$f: \{n \in \mathbb{N} : n \geq n_0\} \rightarrow \mathbb{R}$$

for some fixed integer $n_0 \in \mathbb{N}$.

The sequence associates with the **input variable** n the **output value** a_n .

- The fact that the domain of the function f is the set of natural numbers makes it possible to write the sequence by **listing its values**, in the order in which they follow one another as n increases:

$$a_0, a_1, a_2, \dots, a_n, \dots$$

- The dots after a_n indicate that we are not considering only the first n terms of the sequence (i.e., a **finite set** of numbers), but the whole sequence of *infinitely many terms* (i.e., an **infinite set** of numbers).

Example 1: Sequences

$n \in \mathbb{N}$	$n \mapsto n^2$	$0, 1, 4, 9, 16, \dots$
$n \in \mathbb{N}$	$n \mapsto (-1)^n$	$1, -1, 1, -1, 1, \dots$
$n \in \mathbb{N}$	$n \mapsto a \in \mathbb{R}$	$a, a, a, a, a, \dots$

The last sequence is called a **constant sequence**.

Example 2: Sequences

$$n \in \mathbb{N}, n \geq 1$$

$$n \mapsto \frac{1}{n}$$

$$1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}, \dots$$

$$n \in \mathbb{N}, n \geq 2$$

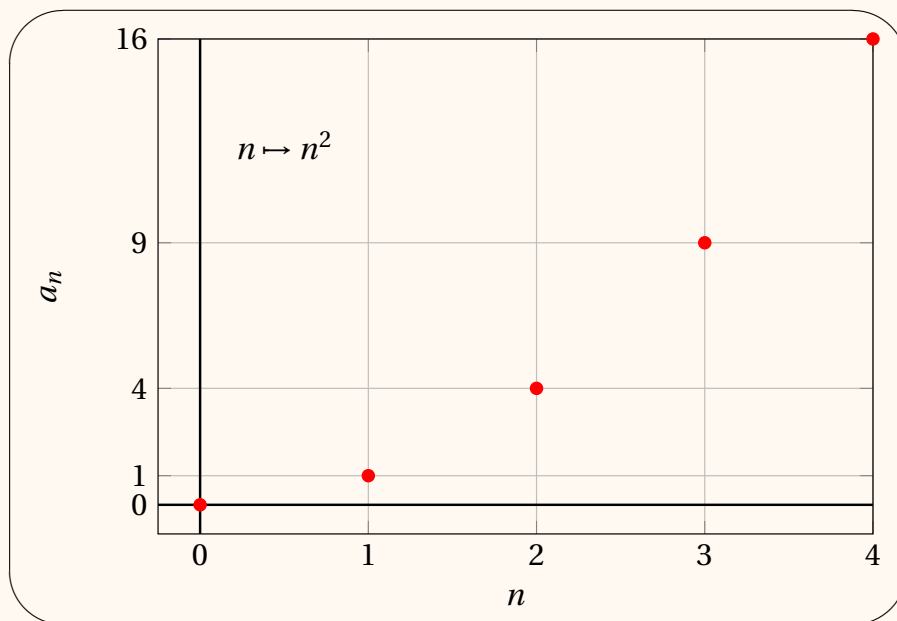
$$n \mapsto \frac{n+1}{n-1}$$

$$3, 2, \frac{5}{3}, \frac{6}{4}, \frac{7}{5}, \dots$$

We can represent sequences graphically by the **points** of the Cartesian plane with coordinates (n, a_n) .

Example 3: Graph of a sequence

The graph of the sequence $n \mapsto n^2$ with $n \in \{0, 1, 2, 3, 4\}$ is:



To denote a sequence we will use the notation:

$$\{a_n\} \quad \text{or} \quad n \mapsto a_n$$

possibly specifying the set in which the input variable n varies (the whole set $\mathbb{N}$ or from a certain value $n_0 \in \mathbb{N}$ onward).

A sequence $\{a_n\}$ is **bounded** if there exist two numbers $m \in \mathbb{R}$ and $M \in \mathbb{R}$ such that:

$$m \leq a_n \leq M, \quad \forall n \in \mathbb{N}$$

It is **bounded below** if m exists. It is **bounded above** if M exists.

Example 4: Bounded sequences

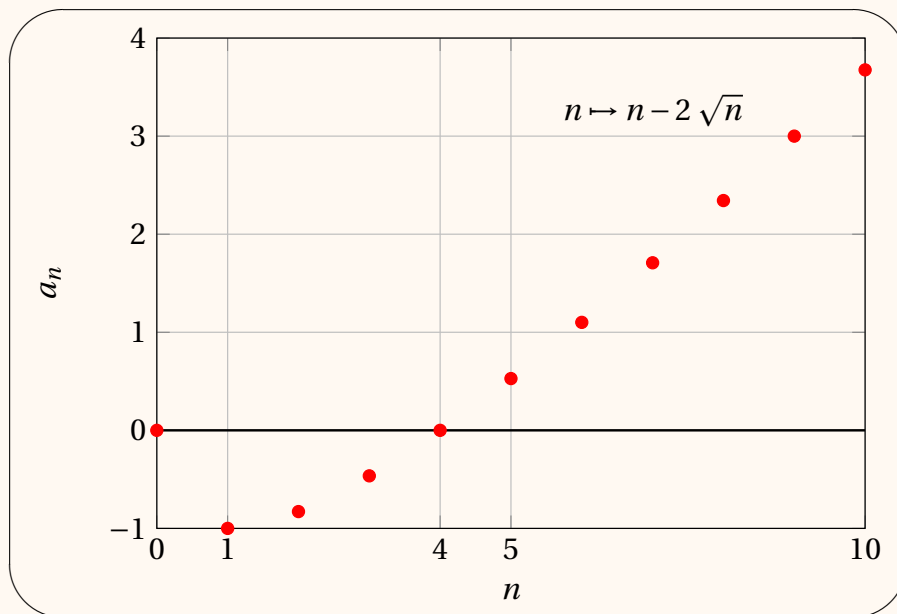
- the sequence $\{(-1)^n\}$ is bounded
- the sequence $\{n^2\}$ is only bounded below
- the sequence $\{(-2)^n\}$ is not bounded (neither below nor above).

Definition 2: Property that holds eventually

We say that a sequence $\{a_n\}$ has (or acquires) a certain property **eventually** if there exists $\tilde{n} \in \mathbb{N}$ such that a_n satisfies that property for every $n \geq \tilde{n}$.

Example 5: Properties that hold eventually

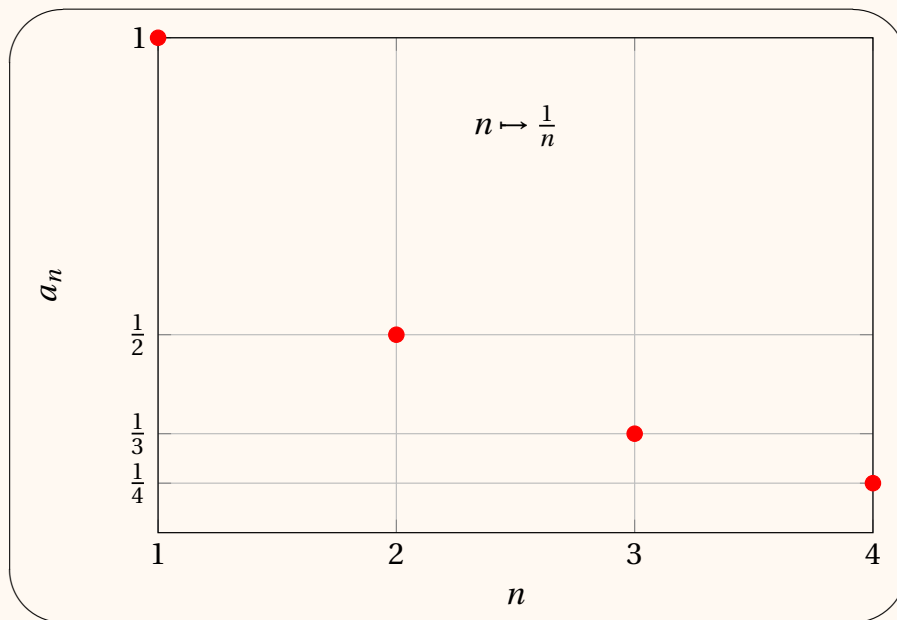
Consider the sequence $\{n - 2\sqrt{n}\}$. The graph of the sequence with $n \in \{0, 1, 2, \dots, 10\}$ is:



This sequence is eventually positive. With $n = 4$ we have $a_n = 0$, hence taking $\tilde{n} = 5$, we have $a_n > 0$ for $n \geq \tilde{n}$.

Example 6: Properties that hold eventually

Now consider the sequence $\{\frac{1}{n}\}$. The graph of the sequence with $n \in \{1, 2, \dots, 4\}$ is:



This sequence is eventually less than 10^{-100} . With $n = 10^{100}$ we have $a_n = 10^{-100}$, hence taking for example $\tilde{n} = 10^{100} + 1$ we have $a_n < 10^{-100}$ for $n \geq \tilde{n}$.

1.1 Convergent sequences and definition of the limit of a sequence

Definition 3: Convergent sequence

A sequence $\{a_n\}$ is called **convergent** if there exists a number $\ell \in \mathbb{R}$ such that:

$$|a_n - \ell| < \varepsilon, \quad \text{eventually} \quad (1)$$

for every $\varepsilon > 0$.

A sequence $\{a_n\}$ is therefore called **convergent** if for every $\varepsilon > 0$ (as small as we like) there exists a number $n(\varepsilon) \in \mathbb{N}$ such that:

$$|a_n - \ell| < \varepsilon \quad \text{for every } n \geq n(\varepsilon)$$

The number $n(\varepsilon)$ depends (in general) on the value of ε . If the sequence $\{a_n\}$ is convergent, then the number $\ell \in \mathbb{R}$ is associated with it.

Definition 4: Limit of a sequence

The number $\ell \in \mathbb{R}$ appearing in inequality (1) is called the **limit of the sequence** $\{a_n\}$, and we write:

$$\lim_{n \rightarrow +\infty} a_n = \ell \quad \text{or, equivalently} \quad a_n \rightarrow \ell \quad \text{for } n \rightarrow +\infty$$

- read, respectively:

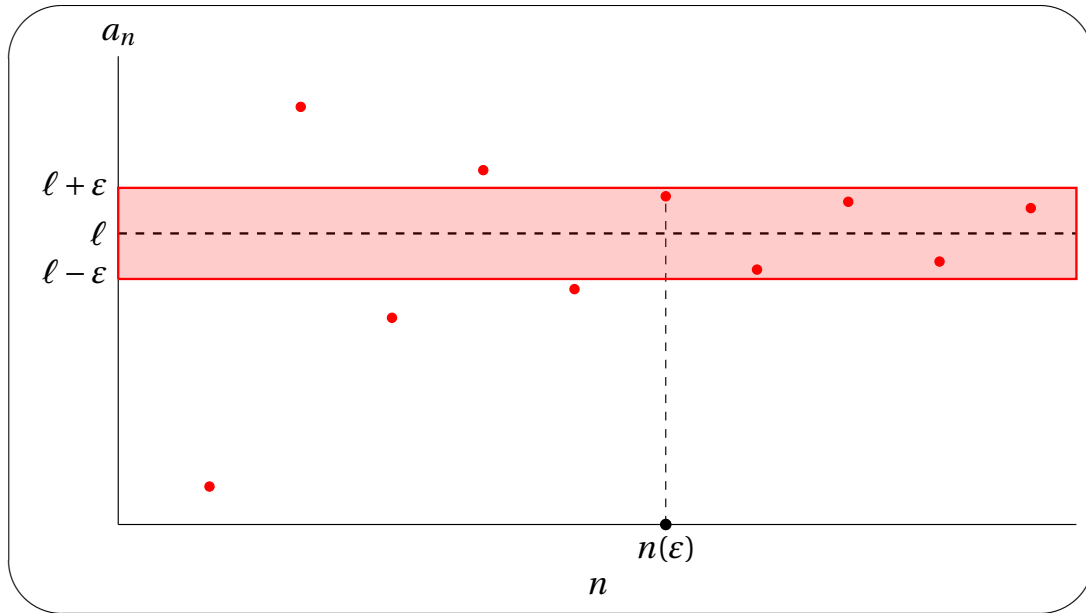
the limit of a_n , as n tends to infinity, is ℓ

a_n tends to ℓ as n tends to infinity

Inequality (1) corresponds to the following two:

$$\ell - \varepsilon < a_n < \ell + \varepsilon \quad (2)$$

- Representing the points of a sequence graphically, we have:



The convergence condition means that, having fixed a horizontal strip “as narrow as we like”:

$$[\ell - \varepsilon, \ell + \varepsilon]$$

from a certain value of n onward, called $n(\varepsilon)$, the points a_n of the sequence no longer leave this strip. In the previous graph, having fixed the width of the strip, the values a_n lie inside the strip for $n \geq n(\varepsilon)$.

Theorem 1: Uniqueness of the limit of a sequence

If a sequence $\{a_n\}$ converges to the limit $\ell \in \mathbb{R}$, then this limit is unique.

Proof. Suppose, by contradiction, that there exist two different limits, ℓ_1 and ℓ_2 , associated with the same sequence $\{a_n\}$. Then, eventually, for every $\varepsilon > 0$ we would have:

$$|\ell_1 - \ell_2| = |\ell_1 - a_n + a_n - \ell_2| \leq \underbrace{|\ell_1 - a_n|}_{=|a_n - \ell_1| < \varepsilon} + \underbrace{|a_n - \ell_2|}_{< \varepsilon} < 2\varepsilon \quad (3)$$

We used the triangle inequality. Since $\varepsilon > 0$ can be chosen arbitrarily small, (3) can be satisfied if and only if:

$$\ell_1 = \ell_2$$

Hence two different values ℓ_1 and ℓ_2 cannot exist and consequently the limit (if it exists) is unique. $\square$

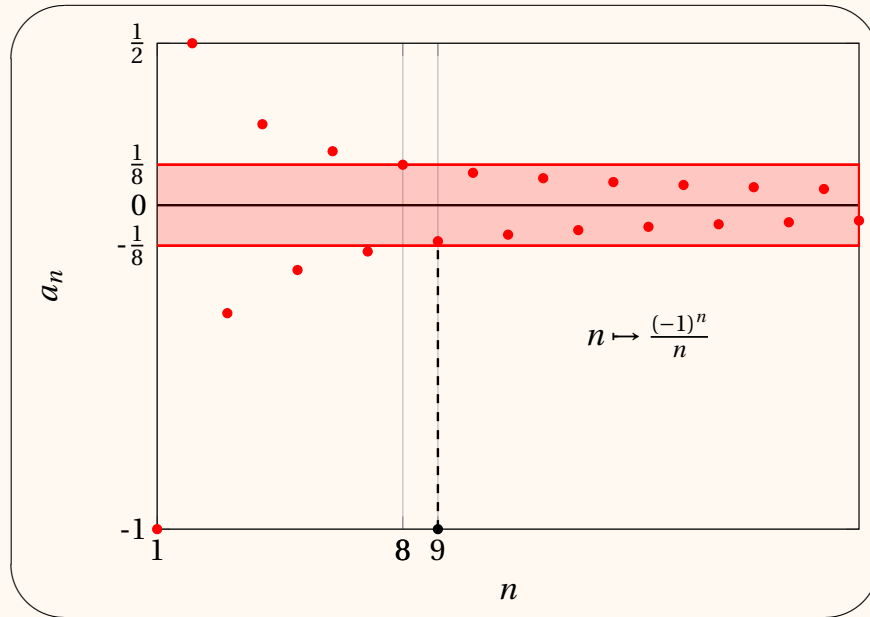
Example 7: Verifying the limit of a sequence

The graph of the sequence $n \mapsto \frac{(-1)^n}{n}$, for example starting from $n = 9$, lies within the horizontal strip:

$$\left[-\frac{1}{8}, \frac{1}{8} \right]$$

given by $\ell = 0$ and $\varepsilon = \frac{1}{8}$. With $n = 8$ we have $a_n = \frac{1}{8}$, with $n = 9$ we have $a_n = -\frac{1}{9}$, hence

$$|a_n| < \frac{1}{8} \quad \text{for every } n \geq 9$$



To prove that:

$$a_n \rightarrow 0 \quad \text{for } n \rightarrow +\infty$$

we must verify that for every $\varepsilon > 0$ there exists $n(\varepsilon) \in \mathbb{N}$ such that

$$n > n(\varepsilon) \Rightarrow |a_n| < \varepsilon$$

The inequality is equivalent to

$$\left| \frac{(-1)^n}{n} \right| < \varepsilon \quad \text{which is satisfied for } n > \frac{1}{\varepsilon}$$

Having fixed $\varepsilon > 0$, it will suffice to choose the first integer

$$n(\varepsilon) > \frac{1}{\varepsilon}$$

to satisfy the condition required by the definition of limit.

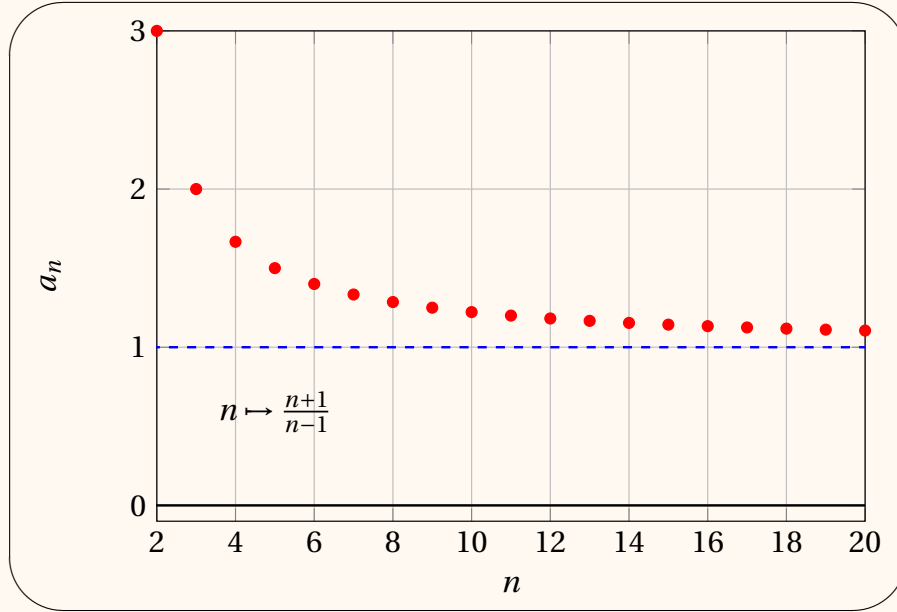
Convergent sequences are (eventually) bounded.

Example 8: Verifying the limit of a sequence

Consider the sequence:

$$n \mapsto \frac{n+1}{n-1} \quad \left(\frac{n+1}{n-1} = \frac{n-1+1+1}{n-1} = 1 + \frac{2}{n-1} \right)$$

This is the graph of the sequence with $n \in \{2, 3, \dots, 20\}$:



We see that the values a_n get closer to 1, hence we try to prove, using the definition of limit, that:

$$\lim_{n \rightarrow +\infty} \frac{n+1}{n-1} = 1$$

We must verify that for every $\varepsilon > 0$ there exists $n(\varepsilon) \in \mathbb{N}$ such that:

$$n > n(\varepsilon) \Rightarrow 1 - \varepsilon < \frac{n+1}{n-1} < 1 + \varepsilon$$

The left inequality is always satisfied (the numerator of the fraction is always larger than the denominator). We take the right one:

$$\frac{n+1}{n-1} - 1 < \varepsilon, \quad \frac{n+1 - (n-1)}{n-1} < \varepsilon, \quad \frac{2}{n-1} < \varepsilon, \quad n-1 > \frac{2}{\varepsilon}$$

$$\text{hence it is satisfied if } n > \frac{2+\varepsilon}{\varepsilon}$$

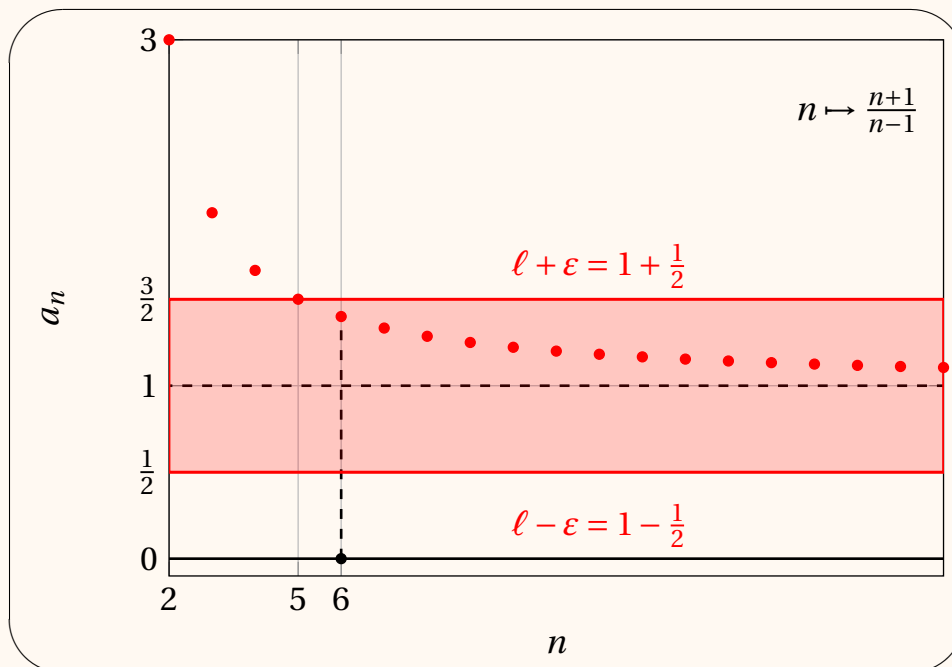
Having fixed $\varepsilon > 0$, it will suffice to choose the first integer

$$n(\varepsilon) > \frac{2+\varepsilon}{\varepsilon}$$

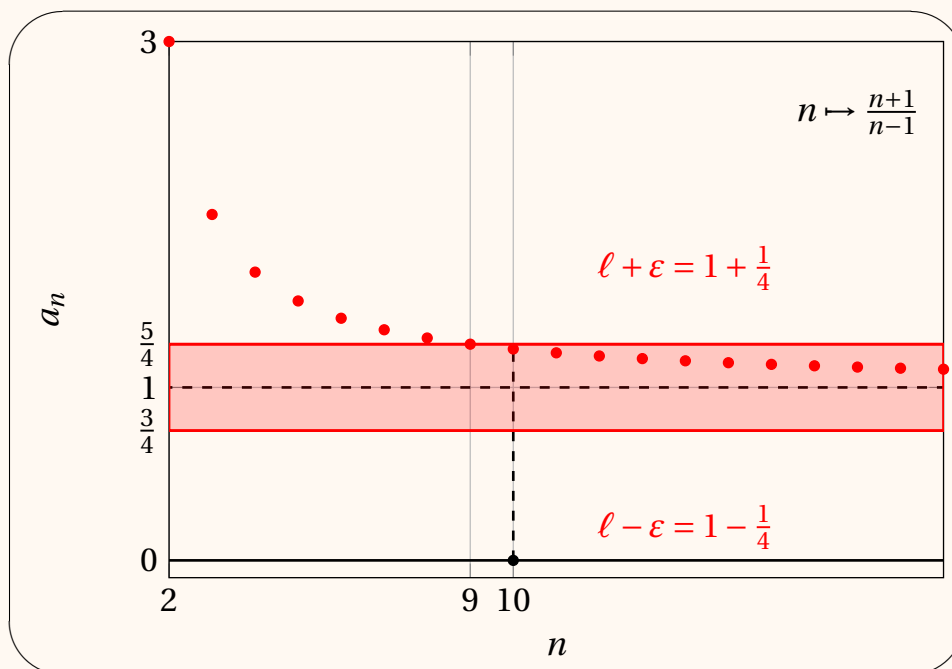
to satisfy the condition required by the definition of limit.

Example 9: Verifying the limit of a sequence

For the previous example, we proved that the limit equals 1. Let us now check what happens when we fix $\varepsilon = \frac{1}{2}$. In this case $n(\frac{1}{2}) > \frac{2+1/2}{1/2} = 5$.



If instead we fix $\varepsilon = \frac{1}{4}$, in this case we have $n(\frac{1}{4}) > \frac{2+1/4}{1/4} = 9$.

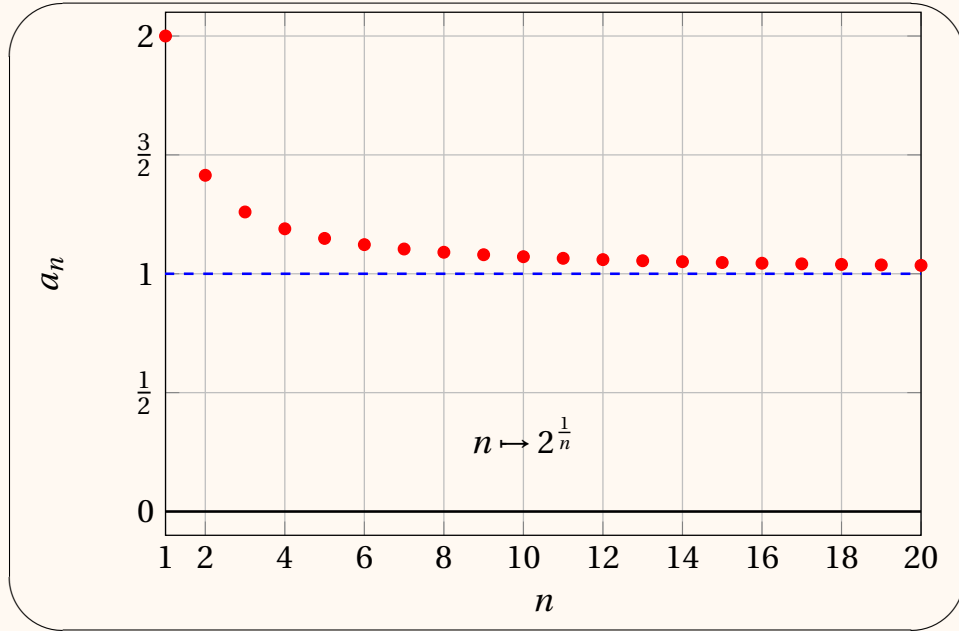


Example 10: Verifying the limit of a sequence

Consider the sequence:

$$n \mapsto 2^{\frac{1}{n}}$$

This is the graph of the sequence with $n \in \{1, 2, \dots, 20\}$:



We see that the values a_n get closer to 1, hence we try to prove, using the definition of limit, that:

$$\lim_{n \rightarrow +\infty} 2^{\frac{1}{n}} = 1$$

We must verify that for every $\varepsilon > 0$ there exists $n(\varepsilon) \in \mathbb{N}$ such that:

$$n > n(\varepsilon) \Rightarrow 1 - \varepsilon < 2^{\frac{1}{n}} < 1 + \varepsilon.$$

The left inequality is always satisfied (2 raised to a positive rational number), while for the right one, taking the logarithm to base 2, we obtain:

$$\frac{1}{n} < \log_2(1 + \varepsilon)$$

Hence it is satisfied if:

$$n > \frac{1}{\log_2(1 + \varepsilon)}$$

Having fixed $\varepsilon > 0$, it suffices to choose the first integer

$$n(\varepsilon) > \frac{1}{\log_2(1 + \varepsilon)}$$

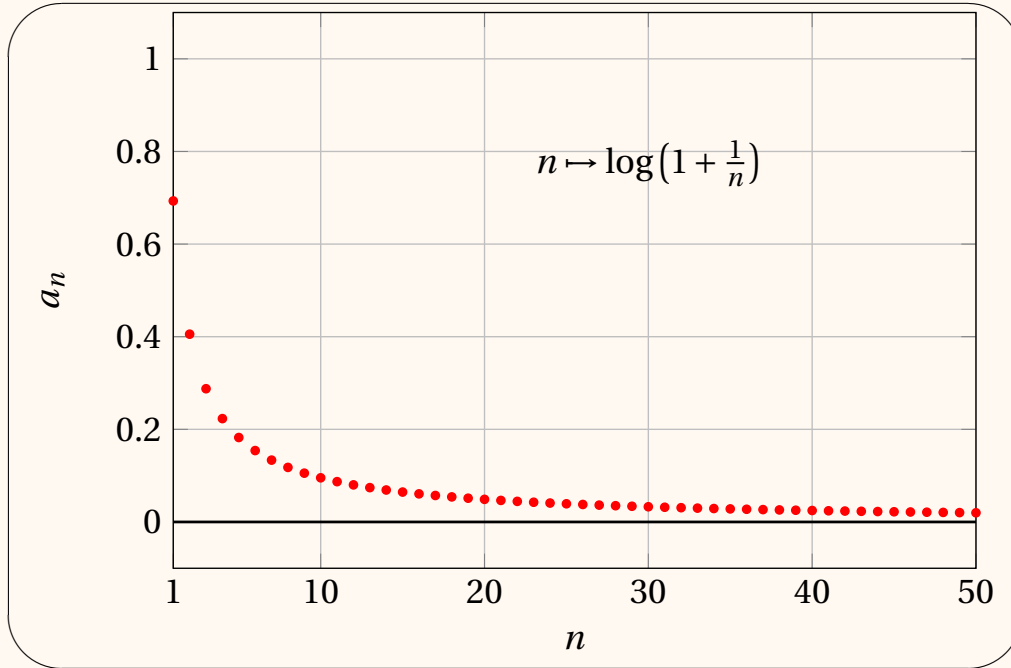
to satisfy the condition required by the definition of limit.

Example 11: Limit of sequences

Consider the sequence:

$$n \mapsto \log\left(1 + \frac{1}{n}\right).$$

This is the graph of the sequence with $n \in \{1, 2, \dots, 50\}$:



We see that the values a_n get closer to 0, hence we try to prove, using the definition of limit, that:

$$\lim_{n \rightarrow +\infty} \log\left(1 + \frac{1}{n}\right) = 0$$

We must verify that for every $\varepsilon > 0$ there exists $n(\varepsilon) \in \mathbb{N}$ such that:

$$n > n(\varepsilon) \Rightarrow -\varepsilon < \log\left(1 + \frac{1}{n}\right) < \varepsilon$$

The left inequality is always satisfied (the logarithm to base e of a number greater than 1), while for the right one, exponentiating, we obtain:

$$\frac{1}{n} + 1 < e^\varepsilon \quad \text{hence it is satisfied if} \quad n > \frac{1}{e^\varepsilon - 1} \quad (e^\varepsilon - 1 > 0 \text{ with } \varepsilon > 0)$$

Having fixed $\varepsilon > 0$, it suffices to choose the first integer

$$n(\varepsilon) > \frac{1}{e^\varepsilon - 1}$$

to satisfy the condition required by the definition of limit.

1.2 Divergent sequences and irregular sequences

Definition 5: Sequence divergent to $+\infty$

A sequence $\{a_n\}$ is called **divergent** to $+\infty$ if for every $M > 0$ there exists a number $n(M) \in \mathbb{N}$ such that:

$$a_n > M \text{ for every } n \geq n(M)$$

Definition 6: Sequence divergent to $-\infty$

A sequence $\{a_n\}$ is called **divergent** to $-\infty$ if for every $M > 0$ there exists a number $n(M) \in \mathbb{N}$ such that:

$$a_n < -M \text{ for every } n \geq n(M)$$

The number $n(M)$ depends (in general) on the value of M .

- In the two cases we will say, respectively, that $+\infty$ and $-\infty$ are the limits of the sequence and we will write, respectively:

$$\lim_{n \rightarrow +\infty} a_n = +\infty \quad \text{or} \quad \lim_{n \rightarrow +\infty} a_n = -\infty$$

The values of a sequence divergent to $+\infty$ eventually exceed any fixed real number. The values of a sequence divergent to $-\infty$ eventually fall below any fixed real number.

The symbols $+\infty$ and $-\infty$ are not numbers.

- If we represent the real numbers on the **Euclidean line**, each number corresponds to a point and each point to a number.
- With the symbols $+\infty$ and $-\infty$ we agree to denote two “points”:

$+\infty$ lies to the *right* of every point of $\mathbb{R}$

$-\infty$ lies to the *left* of every point of $\mathbb{R}$

however, no number corresponds to these two points.

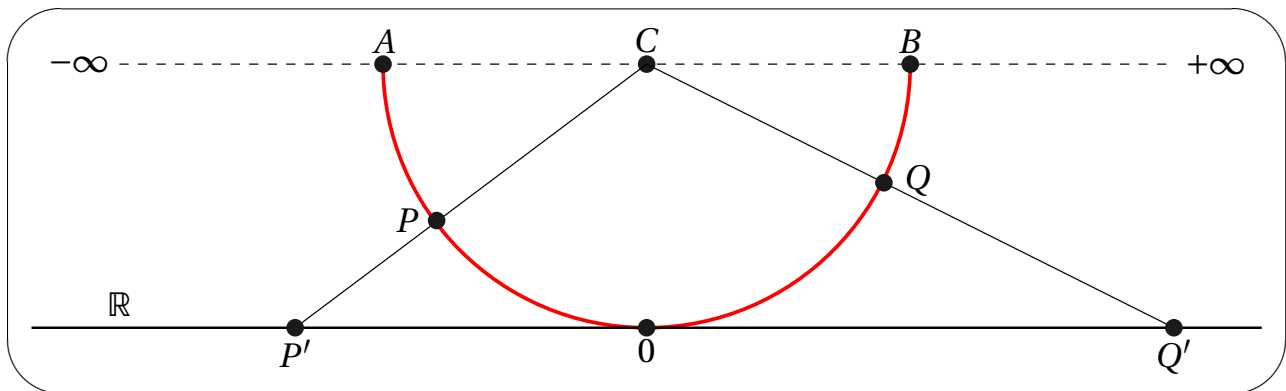
- On the symbols $+\infty$ and $-\infty$ the operations of sum and product with the properties stated in R_1 and R_2 are not defined, even though we will be able to perform these operations “partially” (as we will see later).

Definition 7: The set $\mathbb{R}^*$

The set of real numbers $\mathbb{R}$ with the addition of the two elements $+\infty$ and $-\infty$ will be denoted by:

$$\mathbb{R}^* = \mathbb{R} \cup \{+\infty\} \cup \{-\infty\}$$

- We can represent the set $\mathbb{R}^*$ “visually” by putting the points of the **line** in one-to-one correspondence with those of a **semicircle** (projecting them from the center of the semicircle onto the line $\mathbb{R}$):



- No point on $\mathbb{R}$ corresponds to the points A and B ; we will say that $-\infty$ is the “counterpart” of the point A and $+\infty$ the “counterpart” of B .
- The limit operation becomes *fully meaningful* if set in $\mathbb{R}^*$ instead of $\mathbb{R}$, i.e., the limit of a sequence can be a real number ℓ or $+\infty$ or $-\infty$.

Sequences whose limit is a real number are **convergent**; those whose limit is $+\infty$ or $-\infty$ are **divergent**.

Example 12: Convergent and divergent sequences

- the canonical sequence $\{n\}$ is divergent to $+\infty$;
- the sequence $\{2^n\}$ is divergent to $+\infty$;
- the sequence $\{-2^n\}$ is divergent to $-\infty$;
- the sequence $\{2^{\frac{1}{n}}\}$ is convergent to 1.

Remark 1

$$\lim_{n \rightarrow +\infty} n^\alpha = \begin{cases} +\infty & \text{if } \alpha > 0 \\ 1 & \text{if } \alpha = 0 \\ 0 & \text{if } \alpha < 0 \end{cases}$$

Proof. If $\alpha > 0$, we must prove that the sequence diverges to $+\infty$. We must therefore verify that for every $M > 0$ there exists $n(M) \in \mathbb{N}$ such that

$$n > n(M) \Rightarrow a_n > M$$

The inequality

$$n^\alpha > M \text{ is satisfied for } n > M^{\frac{1}{\alpha}}$$

Hence, having fixed $M > 0$, it will suffice to choose the first integer

$$n(M) > M^{\frac{1}{\alpha}}$$

to satisfy the required divergence condition.

Hence there is no $M > 0$ such that $n^\alpha \leq M$ for $n \in \mathbb{N}$, and the sequence is not bounded. Hence $n^\alpha \rightarrow +\infty$ for $n \rightarrow +\infty$.

If $\alpha < 0$, we must verify that for every $\varepsilon > 0$ there exists $n(\varepsilon) \in \mathbb{N}$ such that

$$n > n(\varepsilon) \Rightarrow -\varepsilon < n^\alpha < \varepsilon$$

The left inequality is always satisfied. For the right inequality, since $n^\alpha = \frac{1}{n^{|\alpha|}}$, we obtain

$$\frac{1}{n^{|\alpha|}} < \varepsilon$$

which is satisfied if

$$n^{|\alpha|} > \frac{1}{\varepsilon} \text{ i.e., for } n > \frac{1}{\varepsilon^{1/|\alpha|}}$$

Having fixed $\varepsilon > 0$, it will suffice to choose the first integer

$$n(\varepsilon) > \frac{1}{\varepsilon^{1/|\alpha|}}$$

to satisfy the condition required by the definition of limit.

□

- Finally, we observe that there are sequences that are neither convergent nor divergent

Definition 8: Irregular (indeterminate) sequence

A sequence that is neither convergent nor divergent is called **irregular** (oscillating) or **indeterminate**.

Example 13: Irregular sequences

- the sequence $\{(-1)^n\}$ is neither convergent nor divergent (but it is bounded)
 - the sequence $\{(-2)^n\}$ is neither convergent nor divergent (and not even bounded).
- For **irregular** sequences, the limit operation is not defined, i.e., their **limit does not exist**.

Summarizing, the operation of computing the limit allows us to answer rigorously the question: **how do the numbers a_n behave as n becomes larger and larger?**

2 Unbounded sets and their suprema/infima

- It is convenient to adopt the convention introduced for limits also for the sup and the inf, extending the definition of these quantities as follows

Definition 9: Supremum and infimum sup and inf (unbounded sets)

If a set $E \subseteq \mathbb{R}$ is not bounded above (below), we will say that

$$\sup E = +\infty \quad (\inf E = -\infty)$$

- In this way, property R_4 of the real numbers can be stated as follows:

$R_4 \rightarrow$ every non-empty set $E \subseteq \mathbb{R}$ has a supremum and an infimum; $\sup E$ ($\inf E$) is a number if E is bounded above (below), otherwise it is $+\infty$ ($-\infty$).

3 Infinitesimal and infinite sequences

Definition 10: Infinitesimal sequence

A sequence $\{a_n\}$ tending to zero is called **infinitesimal**

Example 14: Infinitesimal sequences

- the sequence $\{\frac{1}{n}\}$ is infinitesimal
- the sequence $\{\frac{1}{n^2}\}$ is infinitesimal

- The concept of *infinitesimal* plays a central role and is also fundamental for having a correct and effective **intuitive picture** of the concepts of *infinitesimal calculus*.

an “**infinitesimal**” is not an “infinitely small number” (a meaningless concept) but a **variable quantity** (a sequence or, as we will see, a function) that **becomes indefinitely small**.

Definition 11: Infinite sequence

A sequence $\{a_n\}$ tending to $\pm\infty$ is called **infinite** (an infinity)

Example 15: Infinite sequences

- the sequence $\{n^2\}$ is infinite
- the sequence $\{n!\}$ is infinite

- Sometimes it is possible to specify whether a convergent sequence **approaches its limit from above or from below**

Definition 12: Limit from above

We say that the sequence $\{a_n\}$ tends to $\ell \in \mathbb{R}$ **from above** and we write

$$\lim_{n \rightarrow +\infty} a_n = \ell^+$$

if for every $\varepsilon > 0$ we have that

$$0 \leq a_n - \ell < \varepsilon, \quad \text{eventually.}$$

Definition 13: Limit from below

We say that the sequence $\{a_n\}$ tends to $\ell \in \mathbb{R}$ **from below** and we write

$$\lim_{n \rightarrow +\infty} a_n = \ell^-$$

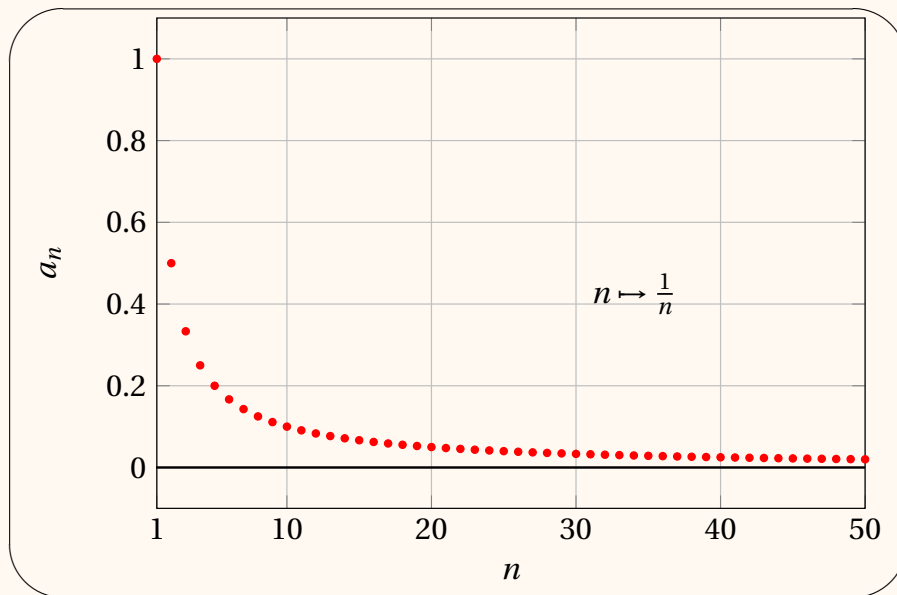
if for every $\varepsilon > 0$ we have that

$$0 \leq \ell - a_n < \varepsilon, \quad \text{eventually.}$$

- Saying that $a_n \rightarrow \ell^+$ for $n \rightarrow +\infty$ means stating that $a_n \rightarrow \ell$ and moreover $a_n \geq \ell$ eventually; hence a_n approaches ℓ *from above*, i.e., it approximates ℓ from above.
- Saying that $a_n \rightarrow \ell^-$ for $n \rightarrow +\infty$ means stating that $a_n \rightarrow \ell$ and moreover $a_n \leq \ell$ eventually; hence a_n approaches ℓ *from below*, i.e., it approximates ℓ from below.

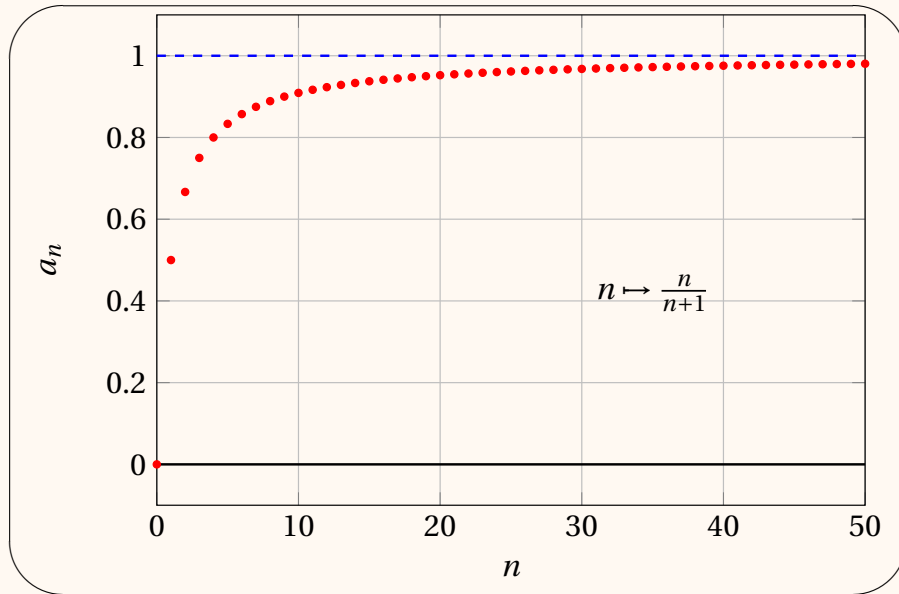
Example 16: Sequence with limit from above

$$\lim_{n \rightarrow +\infty} \frac{1}{n} = 0^+$$



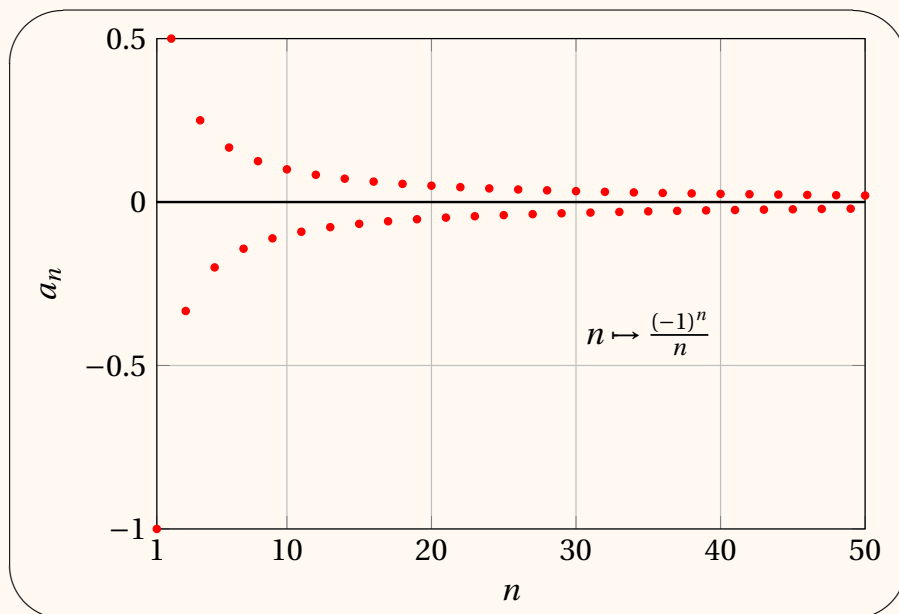
Example 17: Sequence with limit from below

$$\lim_{n \rightarrow +\infty} \frac{n}{n+1} = 1^- \quad \left(\frac{n}{n+1} = \frac{n+1-1}{n+1} = 1 - \frac{1}{n+1} \right)$$



Example 18: Sequence with a limit, but neither from below nor from above

$$\lim_{n \rightarrow +\infty} \frac{(-1)^n}{n} = 0$$



In this case we can state neither that $a_n \rightarrow 0^+$ nor that $a_n \rightarrow 0^-$ for $n \rightarrow +\infty$.

4 Monotone sequences

Definition 14: Monotone sequences

A sequence $\{a_n\}$ is called **monotone increasing** if:

$$a_n \leq a_{n+1}, \forall n$$

A sequence $\{a_n\}$ is called **strictly monotone increasing** if:

$$a_n < a_{n+1}, \forall n$$

A sequence $\{a_n\}$ is called **monotone decreasing** if:

$$a_n \geq a_{n+1}, \forall n$$

A sequence $\{a_n\}$ is called **strictly monotone decreasing** if:

$$a_n > a_{n+1}, \forall n$$

Example 19: Monotone increasing/decreasing sequences

- The sequence $\{n^2\}$ is strictly monotone increasing
 - The sequence $\{\frac{1}{n}\}$ is strictly monotone decreasing
 - The sequence $\{(-1)^n\}$ is not monotone
 - every constant sequence is monotone (increasing or decreasing, not strictly)
- With regard to the limit operation, these sequences are of particular importance; indeed, they are **never irregular**, but are **convergent or divergent** depending on whether they are **bounded or not**.

Theorem 2: Monotone sequence theorem

- Let $\{a_n\}$ be a **monotone increasing sequence that is bounded above**. Then $\{a_n\}$ is convergent, and its limit equals

$$\sup\{a_n : n \in \mathbb{N}\}.$$

- Let $\{a_n\}$ be a **monotone decreasing sequence that is bounded below**. Then $\{a_n\}$ is convergent, and its limit equals

$$\inf\{a_n : n \in \mathbb{N}\}.$$

Proof. We consider the case of monotone increasing sequences that are bounded above.

Since the sequence is bounded above, the set of values taken by the sequence $\{a_n : n \in \mathbb{N}\}$ is bounded above.

By the *supremum property* enjoyed by $\mathbb{R}$, there therefore exists a finite supremum, which we denote by ℓ :

$$\ell = \sup\{a_n : n \in \mathbb{N}\} \quad \text{and} \quad \ell \in \mathbb{R}.$$

We now need to prove that

$$\lim_{n \rightarrow +\infty} a_n = \ell$$

and hence that for every $\varepsilon > 0$ we have

$$\ell - \varepsilon < a_n < \ell + \varepsilon, \quad \text{eventually.}$$

The second inequality is obvious. For every $n \in \mathbb{N}$ we have

$$a_n \leq \ell \quad (\text{and hence } a_n < \ell + \varepsilon)$$

since ℓ is the supremum of $\{a_n : n \in \mathbb{N}\}$ and hence an upper bound.

By definition of supremum, ℓ is the least of the upper bounds of the set $\{a_n : n \in \mathbb{N}\}$. Therefore, since

$$\ell - \varepsilon < \ell,$$

certainly $\ell - \varepsilon$ is not an upper bound of the set $\{a_n : n \in \mathbb{N}\}$. This means that there exists an $n(\varepsilon) \in \mathbb{N}$ for which

$$a_{n(\varepsilon)} > \ell - \varepsilon.$$

On the other hand, the sequence is monotone increasing, therefore for every $n \geq n(\varepsilon)$ we have $a_n \geq a_{n(\varepsilon)}$. We have thus proved that

$$a_n \geq a_{n(\varepsilon)} > \ell - \varepsilon \quad \text{for every } n \geq n(\varepsilon)$$

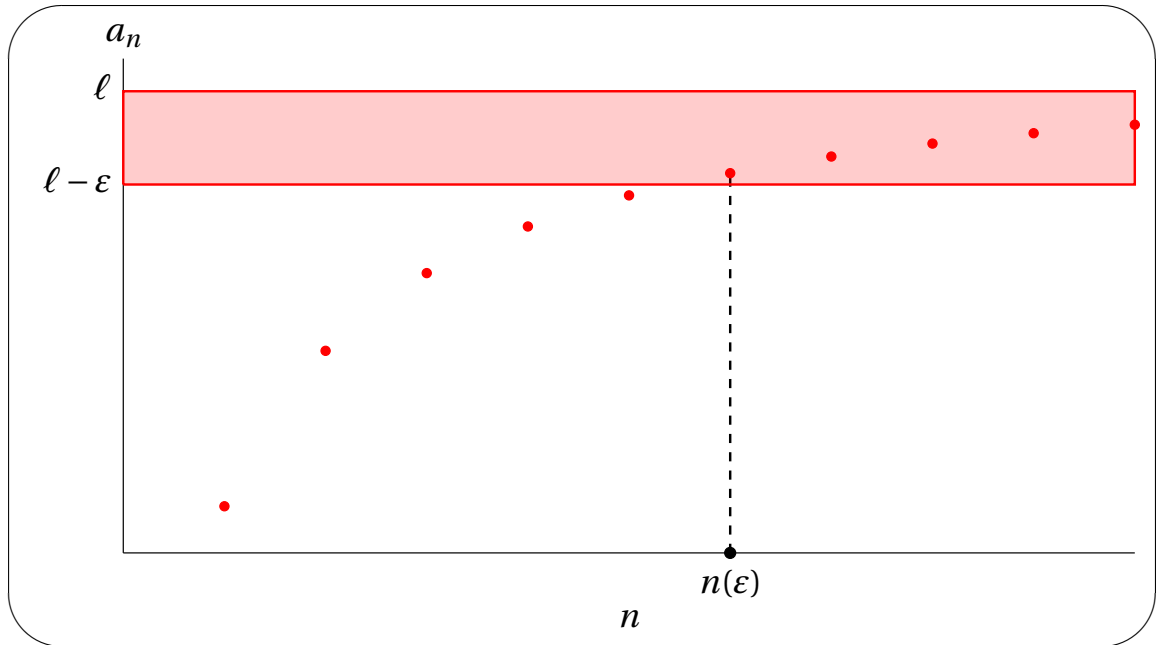
Consequently, we have the claim:

$$\lim_{n \rightarrow +\infty} a_n = \ell$$

The case of monotone decreasing sequences that are bounded below is proved analogously.

□

- The idea of the proof is conveyed by the following figure:



- To express symbolically as well that the limit is the sup (or the inf) of an increasing (or decreasing) sequence, we use the notation

$$a_n \uparrow \ell \text{ or } a_n \downarrow \ell$$

This in particular implies that $a_n \rightarrow \ell^-$ (respectively, ℓ^+), but it contains a further piece of information: the monotonicity of the sequence.

- This theorem is a consequence of the completeness axiom (axiom of continuity) R_4 of the real numbers and therefore holds if the setting we consider is $\mathbb{R}$. For example, it is not true that an increasing and bounded sequence of rational numbers always has a rational limit, i.e., in $\mathbb{Q}$.

Example 20: Increasing and bounded sequence in $\mathbb{Q}$

Let $\{a_n\}$ be the sequence defined as follows:

$$a_0 = 0, a_1 = 0, 1, a_2 = 0, 1011, a_3 = 0, 10110111, a_4 = 0, 1011011101111 \dots$$

At step n we append to the decimal number obtained at the previous step a digit zero followed by n digits equal to 1. The sequence $\{a_n\}$ is evidently increasing, and bounded above (for example, $a_n \leq 1$).

- In $\mathbb{R}$ the sequence converges to the number $\sup\{a_n : n \in \mathbb{N}\}$, which after the decimal point has an infinite, non-periodic decimal expansion (one digit 1, one digit 0, two digits 1, one digit 0, three digits 1, one digit 0, and so on forever)
- Hence the limit of the sequence is an irrational number. This example shows that in the set $\mathbb{Q}$ the monotone sequence theorem is false.

- The monotone sequence theorem can be completed with the next corollary, which considers bounded or unbounded sequences.

Corollary 1: Of the monotone sequence theorem

Let $\{a_n\}$ be a monotone increasing sequence. Then there exists

$$\lim_{n \rightarrow +\infty} a_n = \sup\{a_n : n \in \mathbb{N}\}.$$

Proof. If $\{a_n\}$ is bounded above, the statement is contained in Theorem (??) (monotone sequence theorem).

If instead $\{a_n\}$ is unbounded above, this means that, having fixed $M > 0$, there exists an $n(M) \in \mathbb{N}$ such that

$$a_{n(M)} > M$$

On the other hand, the sequence is increasing, therefore for every $n \geq n(M)$ we have

$$a_n \geq a_{n(M)} > M$$

We have thus proved that for every $M > 0$ we have $a_n > M$, eventually. This means that $a_n \rightarrow +\infty$ for $n \rightarrow +\infty$.

□

Summarizing, we have:

- if $\{a_n\}$ is bounded above, then it converges (and its limit equals the supremum of its values, which in this case is a real number)
- if instead $\{a_n\}$ is unbounded above, then a_n tends to $+\infty$ (which in this case equals the supremum of its values).

A monotone sequence either converges or diverges (it cannot be irregular).

5 Limits of geometric progressions

- Consider the **geometric progression** with common ratio $a \in \mathbb{R}$:

$$1, a, a^2, a^3, \dots, a^n, \dots$$

it is equivalent to the sequence $\{a^n\}$:

$$n \mapsto a^n$$

- If $a > 1$, the sequence is monotone increasing and unbounded above.
- If $a = 1$, the sequence is constant.
- If $0 < a < 1$, the sequence is monotone decreasing and tends to zero.
- If a is negative, the sequence is not monotone.

Remark 2

$$\lim_{n \rightarrow +\infty} a^n = \begin{cases} +\infty & \text{if } a > 1 \\ 1 & \text{if } a = 1 \\ 0 & \text{if } |a| < 1 \\ \text{does not exist} & \text{if } a \leq -1 \end{cases}$$

Proof. If $|a| < 1$, we must verify that for every $\varepsilon > 0$ there exists $n(\varepsilon) \in \mathbb{N}$ such that

$$n > n(\varepsilon) \Rightarrow -\varepsilon < a_n < +\varepsilon$$

Hence we must verify that

$$n > n(\varepsilon) \Rightarrow -\varepsilon < |a|^n < \varepsilon$$

The first inequality is always true, while the second is satisfied if

$$n > \log_{|a|} \varepsilon$$

Having fixed $\varepsilon > 0$, it will suffice to choose the first integer

$$n(\varepsilon) > \log_{|a|} \varepsilon$$

to satisfy the condition required by the definition of limit.

If $a > 1$, we must verify that for every $M > 0$ there exists $n(M) \in \mathbb{N}$ such that

$$n > n(M) \Rightarrow a_n > M$$

The inequality

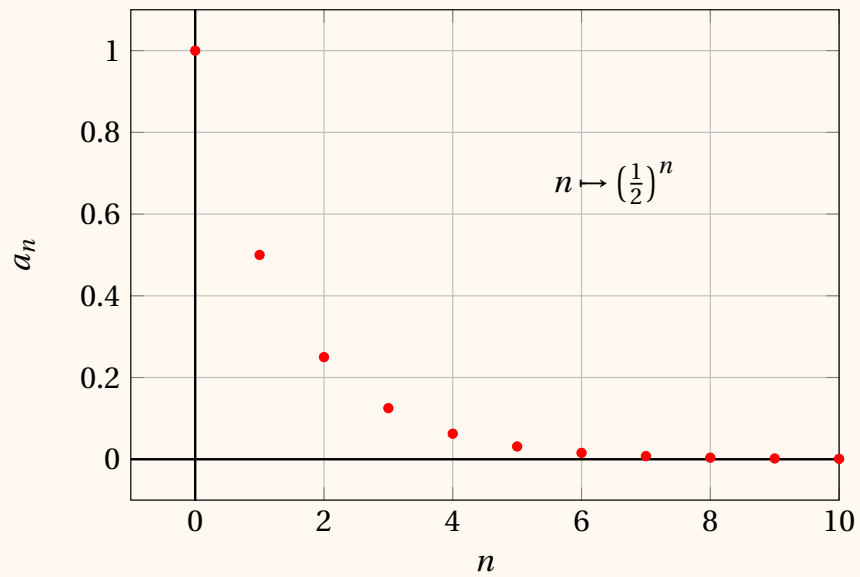
$$a^n > M \text{ is satisfied for } n > \log_a M$$

Hence, having fixed $M > 0$, it will suffice to choose the first integer

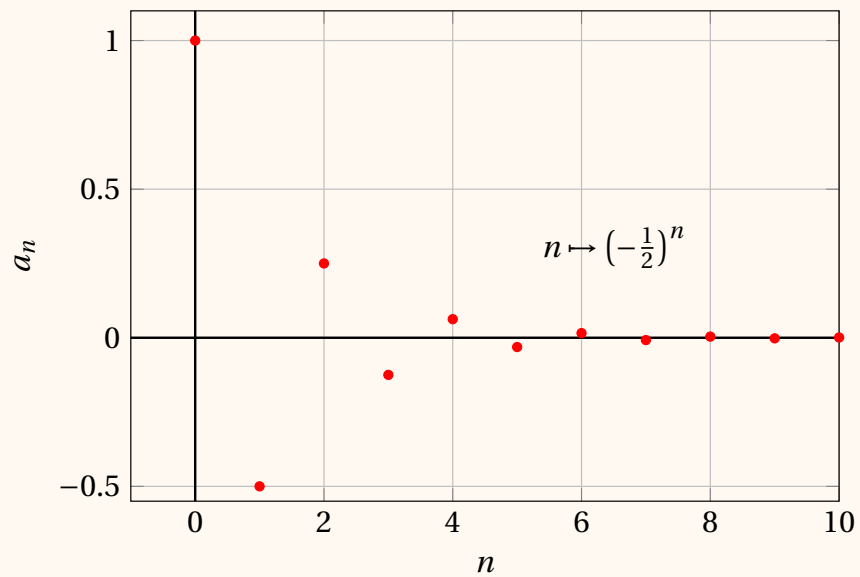
$$n(M) > \log_a M$$

to satisfy the required divergence condition. □

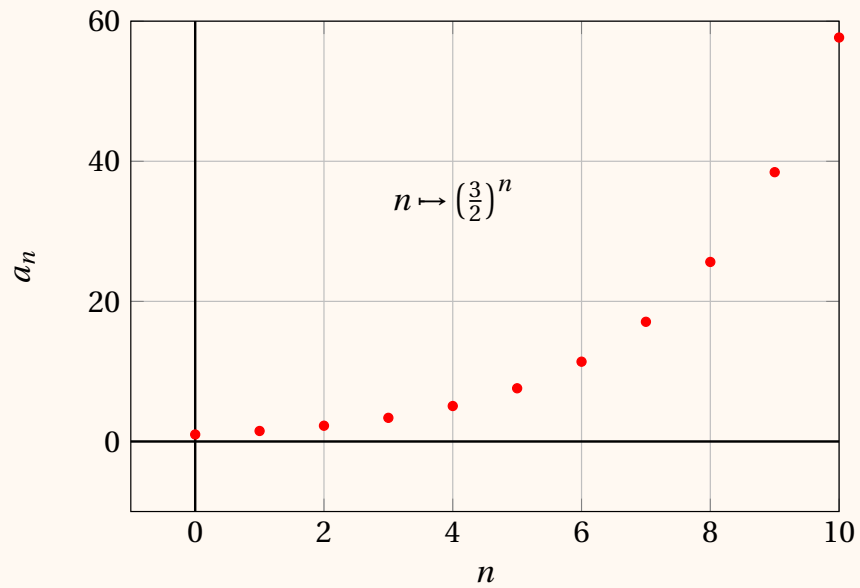
Example 21: Infinitesimal and monotone decreasing geometric progression



Example 22: Infinitesimal (but not monotone) geometric progression



Example 23: Divergent geometric progression



Example 24: Geometric progression that is neither convergent nor divergent

